# DEVESH VARMA

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## SKILLS

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|---|---|
| **Languages** | C, C++, Python |
| **Compilers & Graphics** | LLVM, MLIR, SPIR-V, Vulkan, Graphics Compilers, Compiler Optimization |
| **Tools and Technologies** | Git, Gerrit, CUDA, LaTeX |
| **AI-Assisted Development** | GitHub Copilot, Agentic AI Coding Tools, Automated Compiler-Correctness Test Generation |
| **Other** | Database Systems, Computer Graphics |

## EXPERIENCE

**Samsung Semiconductor** *Bengaluru, India*
*Associate Staff Engineer* *March 2026 - Present*

- Implemented the `VK_EXT_fragment_shader_interlock` Vulkan extension end to end, from the xgl layer down to the compiler layer. Since there was no hardware support, emulated the critical section in software.

*Senior Engineer* *August 2024 - March 2026*

- Increased FPS by 1.76% on the Aztec Normal benchmark for Samsung's AMD RDNA-based Xclipse GPU by optimizing the motion-blur shader.
- Collaborated cross-functionally to debug, profile, and resolve GPU driver performance bottlenecks.

## OPEN SOURCE CONTRIBUTIONS

**SPIRV-Interpreter** *August 2025*

- Fixed the OpSMod instruction and implemented OpSRem per the SPIR-V specification to correct modulo arithmetic behavior, addressing maintainer feedback across multiple review rounds.

**SPIRV-Cross (Khronos Group)** *February 2026*

- Refactored the CLI's help output into categorized flags, replacing a single overwhelming interface.

## PROJECTS

**An MLIR based compiler for Efficient Database Systems** *M.Tech Thesis*
*Mentor: Prof. R. Govindarajan* *May 2023 - June 2024*

- Implemented a GPU-targeted hash-join algorithm in MLIR using the Linalg and SCF dialects, lowering through LLVM IR to GPU kernels, with a nested-loop join baseline for comparison.
- Benchmarked joins on relations of up to 100M rows; an optimized second iteration cut kernel execution time by 25% (2.0s to 1.5s) for a 10M-row join producing over 1 billion output rows, profiled using NVIDIA Nsight.

**Optimized Matrix Multiplication** *Course Project*
*Mentor: Prof. Arkaprava Basu & R. Govindarajan* *Sept - Oct 2022*

- Identified performance bottlenecks in the reference implementation and built optimized single-threaded, multi-threaded, and CUDA versions by exploiting spatial locality and memory prefetching.
- Achieved up to 20x speedup over the single-threaded CPU baseline.

## EDUCATION

**Indian Institute of Science, Bangalore** *August 2022 - June 2024*
*Master of Technology in Computer Science and Engineering* **GPA: 7.7**
*GATE 2022: All India Rank 110 among 77,257 candidates*

**Thakur College of Engineering and Technology**, University of Mumbai *August 2017 - July 2021*
*B.E in Computer Engineering* **GPA: 9.49**